

Complete GitHub Classroom Workflow for Faculty

This document provides a complete step-by-step workflow for using GitHub Classroom starting from creating a classroom to downloading all student submissions for evaluation.

PART 1 — If you do not have a GitHub account then Create GitHub Account otherwise sign

Step 1 — Create GitHub Account

Go to

<https://github.com/signup>

and create a GitHub account for faculty and students.

PART 2 — Install Required Software

Step 2 — Install Git

Ubuntu/Linux:

```
sudo apt update  
sudo apt install git
```

Verify using:

```
git --version
```

Windows:

Download Git from:

<https://git-scm.com/download/win>

Run the installer and keep default settings.

Verify using:

```
git --version
```

macOS:

Install using Homebrew:

```
brew install git
```

Or install Xcode Command Line Tools:

```
xcode-select --install
```

Verify using:

```
git --version
```

Step 3 — Install GitHub CLI

Install GitHub CLI from:

<https://cli.github.com/>

Verify using:

```
gh --version
```

Windows:

Download GitHub CLI from:

<https://cli.github.com/>

Run the installer and complete setup.

Verify using:

```
gh --version
```

macOS:

Install using Homebrew:

```
brew install gh
```

Verify using:

```
gh --version
```

Step 4 — Install GitHub Classroom Extension

Run:

```
gh extension install github/gh-classroom
```

Verify using:

```
gh classroom --help
```

PART 3 — Authenticate GitHub

Step 5 — Login to GitHub

Run:

```
gh auth login --scopes "repo,read:org"
```

Choose:

- GitHub.com
- HTTPS
- Login with browser

Step 6 — Configure Git Authentication

Run:

```
gh auth setup-git
```

Step 7 — Verify Authentication

Run:

```
gh auth status
```

PART 4 — Create GitHub Classroom

Step 8 — Open GitHub Classroom

Open:

```
https://classroom.github.com/
```

Step 9 — Create Organization

Create or select an organization such as VIT-CC.

Step 10 — Create Classroom

Provide classroom name and description.

PART 5 — Create Assignment

Step 11 — Create Assignment

Inside classroom, click 'New Assignment'.

Step 12 — Configure Assignment

Provide assignment title, deadline, and visibility.

Step 13 — Add Starter Repository

Optionally add starter code and README files.

Step 14 — Generate Assignment Link

Share the generated assignment invitation link with students.

PART 6 — Student Submission Workflow

Step 15 — Student Accepts Assignment

Student opens invitation link and accepts assignment.

Step 16 — Student Clones Repository

Student runs:

```
git clone <repo-url>
```

Step 17 — Student Writes Code

Students edit .cpp, .py, .java or other files.

Step 18 — Student Submits

Students commit and push changes:

```
git add .
```

```
git commit -m "submission"
```

```
git push
```

PART 7 — View Student Submissions

Step 19 — Open Assignment Dashboard

Open the assignment dashboard inside GitHub Classroom to view submissions.

PART 8 — Download All Student Submissions

Step 20 — Create Working Folder

Run:

```
mkdir githubCR  
cd githubCR
```

Step 21 — List All Repositories

Run:

```
gh repo list VIT-CC --limit 100
```

Step 22 — Bulk Clone All Assignment Repositories

Run:

```
mkdir submissions  
cd submissions
```

```
for repo in $(gh repo list VIT-CC --limit 100 --json name -q '[][.name' | grep code-  
assignment-1); do  
    echo "Cloning $repo"  
    gh repo clone VIT-CC/$repo  
done
```

PART 9 — Access Student Files

Step 23 — View Downloaded Repositories

Run:

```
ls
```

Step 24 — Find All Code Files

Examples:

```
find . -name "*.cpp"  
find . -name "*.py"  
find . -name "*.java"
```

PART 10 — Collect All Codes into One Folder

Step 25 — Copy All Programs

Run:

```
mkdir ../all_codes
```

```
find . \( -name "*.cpp" -o -name "*.py" -o -name "*.java" \) -exec cp {} ../all_codes/ \;
```

PART 11 — Update Latest Student Submissions

Pull Latest Changes

Run:

```
for dir in */ ; do
  cd "$dir"
  git pull
  cd ..
done
```

PART 12 — AI-Based Evaluation Pipeline

Recommended Workflow

1. Clone repositories
2. Extract code files
3. Run hidden test cases
4. Send code and rubric to GEM/LLM
5. Generate marks and analytics

Useful Links

GitHub

<https://github.com/>

GitHub Classroom

<https://classroom.github.com/>

GitHub CLI

<https://cli.github.com/>

Judge0

<https://judge0.com/>

GitHub REST API

<https://docs.github.com/en/rest>

Common Errors and Fixes

Problem	Solution
Repository not found	Check permissions and organization access
Private repo clone fails	Run <code>gh auth setup-git</code>
gh classroom clone fails	Use <code>gh repo clone loop</code>

Student repo missing

Permission denied

Git authentication failed

Student has not accepted assignment

Accept organization invitation

Re-run gh auth login